

Biometric Smart Lock, Time Wallet & Accounting System

1. Project Overview

This proposal outlines the end-to-end development of an intelligent multi-functional platform that integrates biometric smart lock access, a time-based digital wallet, and a robust accounting system. The target audience includes educational institutions, residential and commercial real estate, event venues, and public businesses requiring controlled access and automated billing.

The proposed system introduces a secure, cloud-powered infrastructure that allows biometric authentication (fingerprint/face), NFC/RFID card support, rechargeable time wallets, real-time dashboards, and advanced accounting—all managed through web and mobile applications.

2. Key Features

Biometric Access Control

- Fingerprint or facial recognition for entry.
- Role-based access for Admin, Staff, Customer, Technician.

Time Wallet System

- Rechargeable wallets using ATM kiosk, mobile money, or cards.

- Configurable daily/hourly billing logic.
- Real-time balance deduction and receipt generation.

RFID/NFC Card Integration

- Smart card linked to user account.
- Offline functionality with auto-sync capability.

School Attendance Automation

- Automated logging of student entry and exit.
- Attendance-linked fee calculation.
- Reports generated daily for administrators and parents.

Real Estate Access Control

- Time-based and flat-rate billing options.
- Real-time access logs and billing summaries per tenant.
- Recharge and extension features available through mobile app.

Events and Venue Access

- Smart ticketing via QR or NFC.
- Real-time wallet deduction for entry and re-entry tracking.

ATM-integrated Kiosks

- Biometric + card access to user wallet.
- On-site card recharge and issuance.
- Supports ads, mobile money, and bank transactions.

Integrated Accounting Suite

- Full ledger tracking for income, expenses, taxes, and loans.
- Auto-generated financial reports: P&L, balance sheet, tax summary.
- Export support for Excel, PDF, and QuickBooks.

Admin Dashboard

- Device health monitoring.
- Multi-branch administration.
- Real-time analytics for usage and revenue.

Notification & Alerts System

- SMS, email, and push alerts for low balance, billing, and access events.

User Portal (Web & Mobile)

- Recharge, view usage and receipts.
- Manage authentication preferences (PIN, biometrics).

3. Technology Stack

Frontend:

- Web: React.js
- Mobile: Flutter (iOS & Android)

Backend:

- Node.js or Python (Django/Flask, depending on business case)

Database:

- PostgreSQL (Primary), Redis (Cache)

Biometric Integration:

- SDK/APIs from ZKTeco, Suprema (or approved third-party vendors)

Cloud Infrastructure:

- AWS / Google Cloud / Azure (based on client preference and locality)

Hardware Integration:

- IoT edge device for biometric locks and kiosks
- NFC/RFID card reader modules

Payment & API Integrations:

- Bank integrations: Interswitch, Paystack, Flutterwave
 - Mobile Money: MTN, Airtel, Safaricom (M-PESA)
 - Notification Services: Twilio, Africa's Talking
 - Tax APIs: FIRS (Nigeria), URA (Uganda), KRA (Kenya)
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4. Roadmap & Milestones

Phase 1 – MVP Development (Weeks 1–4)

- Core backend system
- User portal (basic wallet functions)
- School use-case pilot

Milestone 1: \$750

Phase 2 – Biometric Lock + Wallet Integration (Weeks 5–8)

- Biometric SDK integration
- Time wallet automation
- Card & NFC support

Milestone 2: \$1500

Phase 3 – Real Estate & Events Module (Weeks 9–14)

- Tenant access module
- Event QR/NFC ticketing + time billing

Milestone 3: \$1500

Phase 4 – Kiosk & ATM Interface (Weeks 15–20)

- Kiosk API + Biometric login
- Bank recharge and card management

Milestone 4: \$1000

Phase 5 – Accounting System Development (Weeks 21–23)

- Financial reporting engine
- Tax calculation & export functionality

Milestone 5: \$1000

Phase 6 – Admin Dashboard & Analytics (Weeks 24–27)

- Multi-tenant dashboard
- Device health and transaction monitoring

Milestone 6: \$1000

Phase 7 – Final Testing, Compliance & Localization (Weeks 28–32)

- Regional compliance (GDPR, KRA, URSB, etc.)
 - Language, currency & tax rule settings
 - Final testing & launch prep
- Milestone 7: \$1000**
- **Note:** All milestone payments must be made **before** the milestone begins to ensure timely and uninterrupted development.

5. Budget

The total development cost for the Biometric Smart Lock, Time Wallet & Accounting System is **\$7,750**. This includes software development, system design, QA, integrations, and deployment support.

6. Deployment & Hosting

- Hosting will be managed on client's preferred cloud platform (AWS, GCP, or Azure).
 - Full deployment to production environments with SSL and domain configuration.
 - Devices (biometric locks, kiosks) require on-site or remote pairing; hardware costs not included.
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7. Maintenance & Support

- **Post-Launch Support (6 months free):**
Includes bug fixes, updates, and emergency support.
- **Extended Maintenance Plan (Optional):**
Annual maintenance contract for continued support, server monitoring, and feature updates.
- **Hardware SLA (Optional Tier):**
On-site or remote support for connected devices and kiosks.

8. Additional Considerations

- Hardware procurement and installation (biometric locks, NFC cards, kiosks) are **not included** in this proposal and may be provided separately.
- Licensing costs for biometric SDKs will be determined based on the selected vendor.
- Custom branding (logos, themes) will be provided as part of the final phase.
- Compliance audits may be required depending on country of deployment.

9. Conclusion

We believe this powerful, integrated system will deliver immense value across multiple industries, automating access control, billing, and financial tracking with a seamless user experience.

Should you have any questions or require adjustments to fit your exact business model, we are happy to customize this offering.

Looking forward to collaborating with you on this impactful project.

Best Regards,

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